

Deployment, Configuration and Ingestion

“Patient Journey Intelligence”

Nabin Khadka

Lead Engineer, JohnSnowLabs

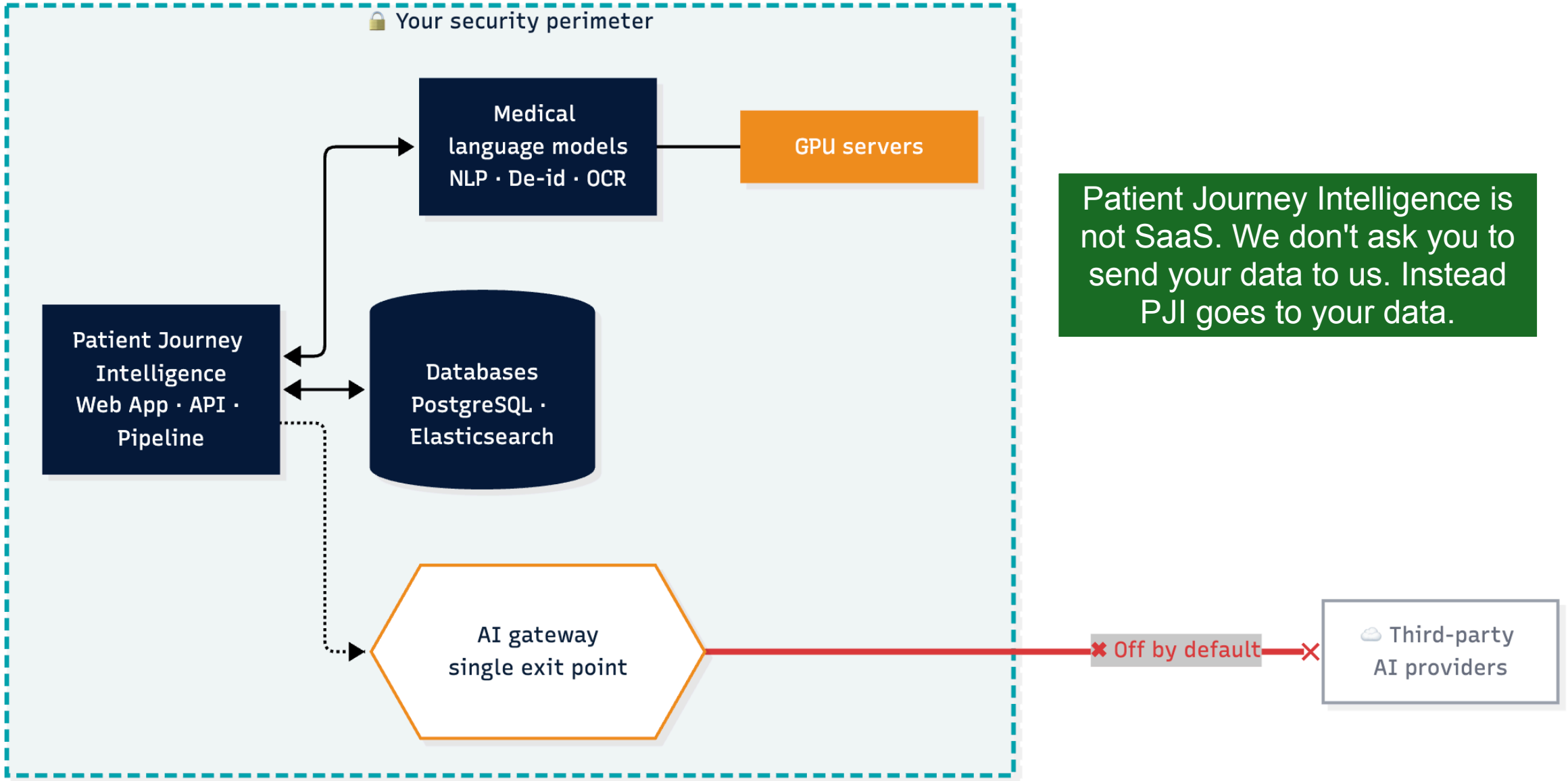
Agenda

1. Deploy anywhere, keep PHI at home (20 min)
2. Connect every source (15 min)
3. Ingestion jobs, more than copying files (25 min)
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6. Data out: export, FHIR and MCP (8 min)

Where does your patient data live right now?

EVERYWHERE

Where are you allowed to process it?



Patient Journey Intelligence is not SaaS. We don't ask you to send your data to us. Instead PJI goes to your data.

3 promises

1: Your clinical **data never leaves** your environment. PJI runs inside your network, your VPC, or your data center.

2: The **medical language models run locally**. Extraction, de-identification and terminology mapping all run on infrastructure you control. Nothing is sent to an outside model provider by default.

3: All external AI service calls go through **a single gateway** service. No other component holds provider credentials or calls an outside AI endpoint. For your security team, that means a single place to **audit, restrict or switch off**.

On top of that you get the basics your compliance team expects:

- Encryption at rest and in transit.
- Single sign-on and role-based access.
- Audit logs for user, permission and login activity.

Deploy anywhere, keep PHI at home

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The Release Kit



3 core scripts

- `k3s-installer.sh` – bootstrap K3s + Helm + kubectl
- `pj-app-installer.sh` – first install
- `pj-app-updater.sh` – safe upgrade

Required Files
(in same folder)

- pj-app-<version>.tgz
- license.json

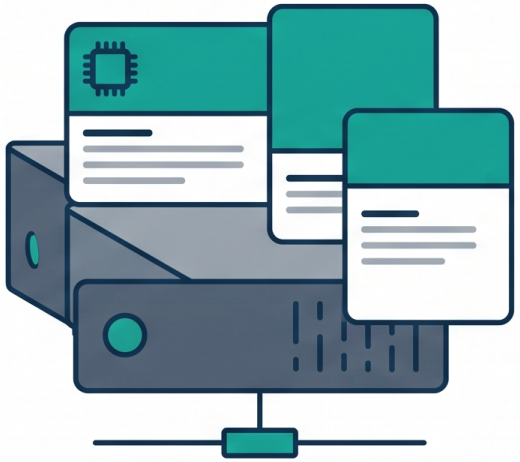
⚠ Scripts exit if a file is missing

Getting familiar with main scripts and necessary files

Deploy anywhere, keep PHI at home

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K3s: Single-Node Linux Host



Minimum: 16 vCPU / 64 GiB – ingestion
Full stack: 32 vCPU / 128 GiB

1. `sudo ./k3s-installer.sh`

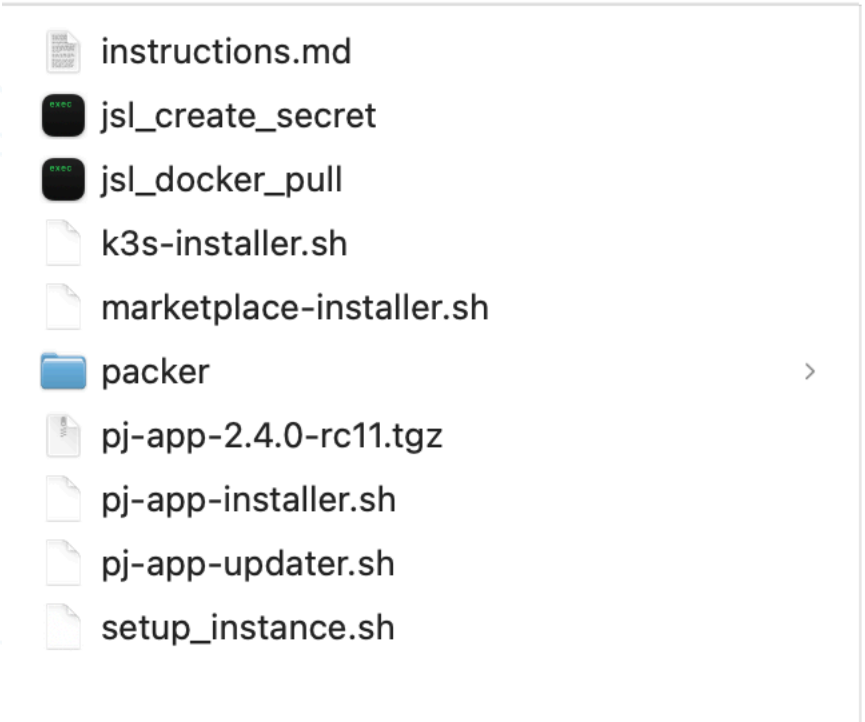
2.
```
sudo ./pj-app-installer.sh
  --namespace dev
  --host pjapp.example.com
```

Multi-node? Add `--nfs` for shared pipeline cache

Simplest setup for getting started!

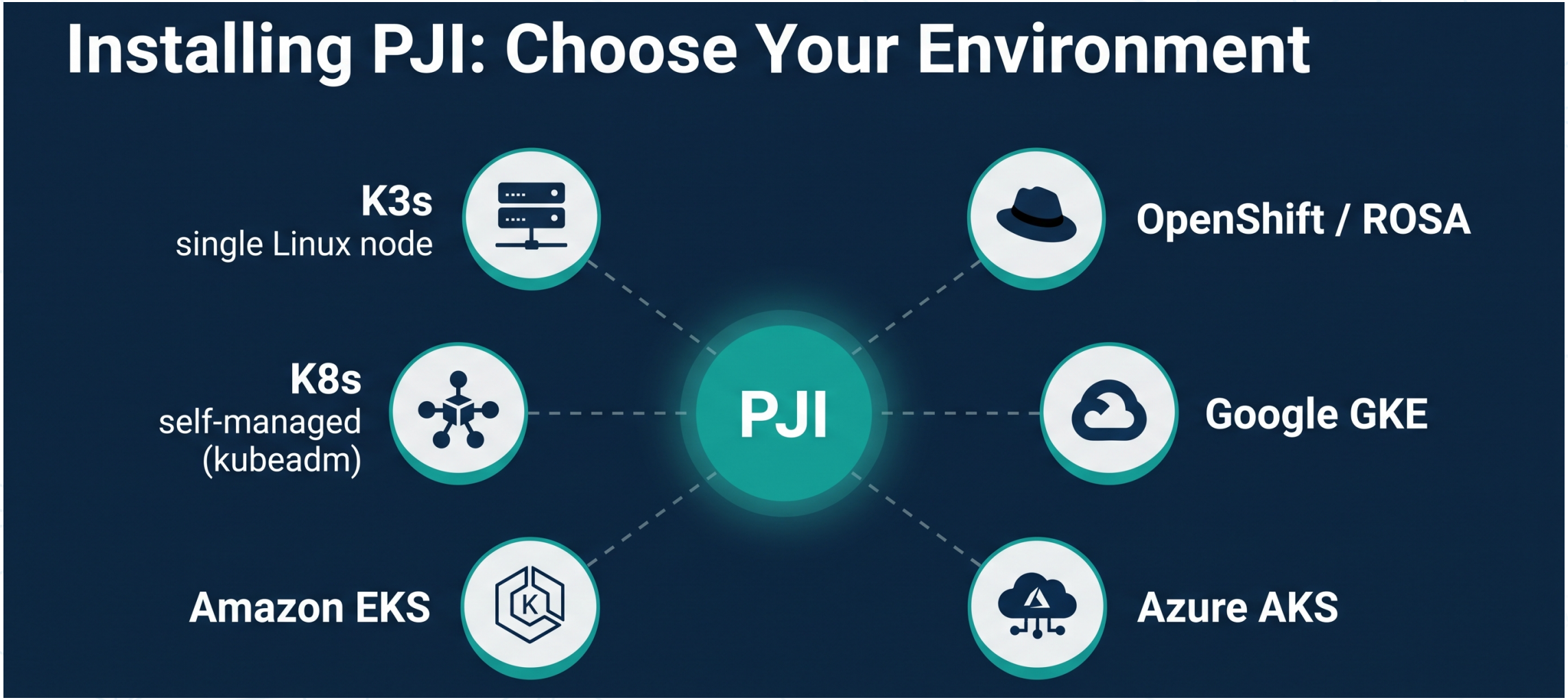
More on installation support

- instructions.md
- https://www.johnsnowlabs.com/pji/installation/overview
- HTTPs, your certificates, your domain.
- Optionally use managed services. Eg. Amazon RDS can replace the bundled PostgreSQL and Amazon OpenSearch can replace the bundled Elasticsearch.
- On the data side, PJI reads from S3, AWS HealthLake for FHIR data, and AWS HealthImaging for medical images.



license.json file is required in this location to start the installation

Deploy anywhere, keep PHI at home



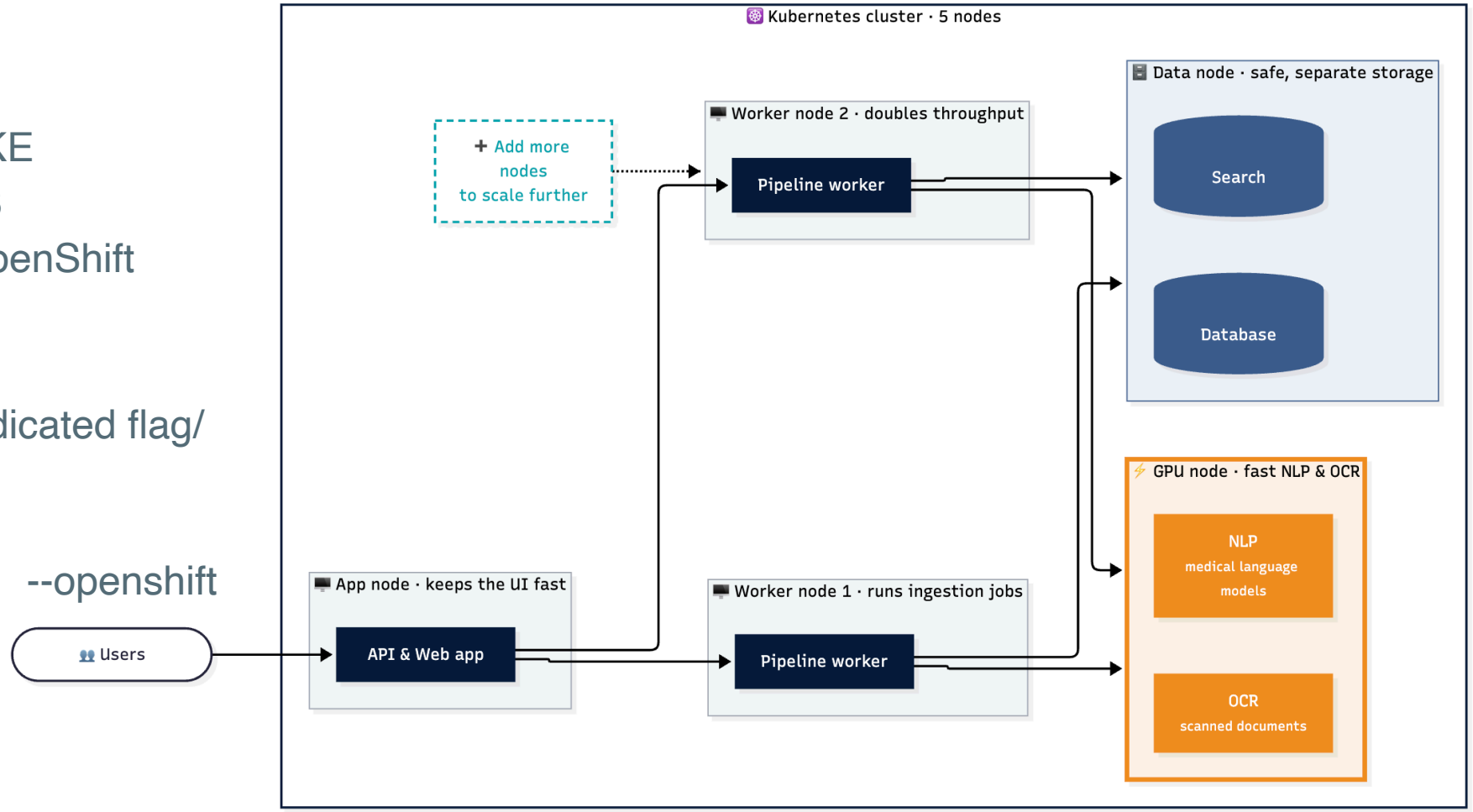
Seamless deployment in multiple platforms and service providers!

Multi-node cluster

- AWS EKS
- Google GKE
- Azure AKS
- RedHat OpenShift

All with a simple dedicated flag/ argument

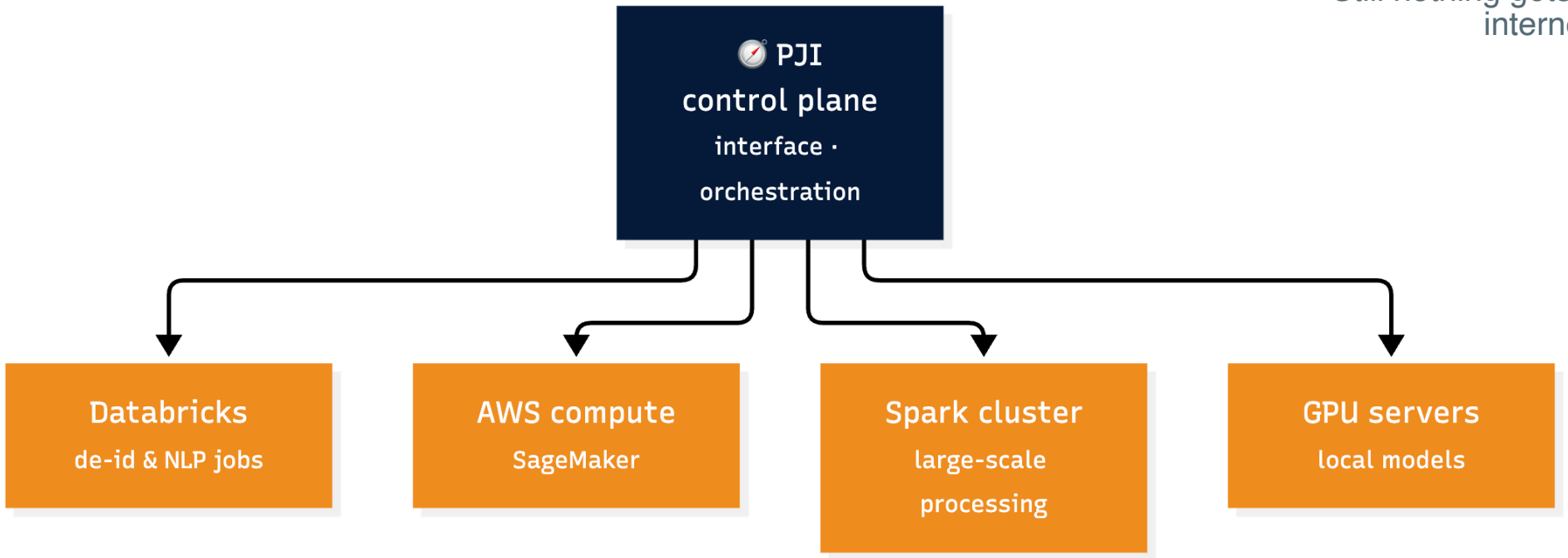
—eks --gke --aks --openshift --k8s



Hybrid: Use what you already own

Most common yet complicated part, especially for large organizations.

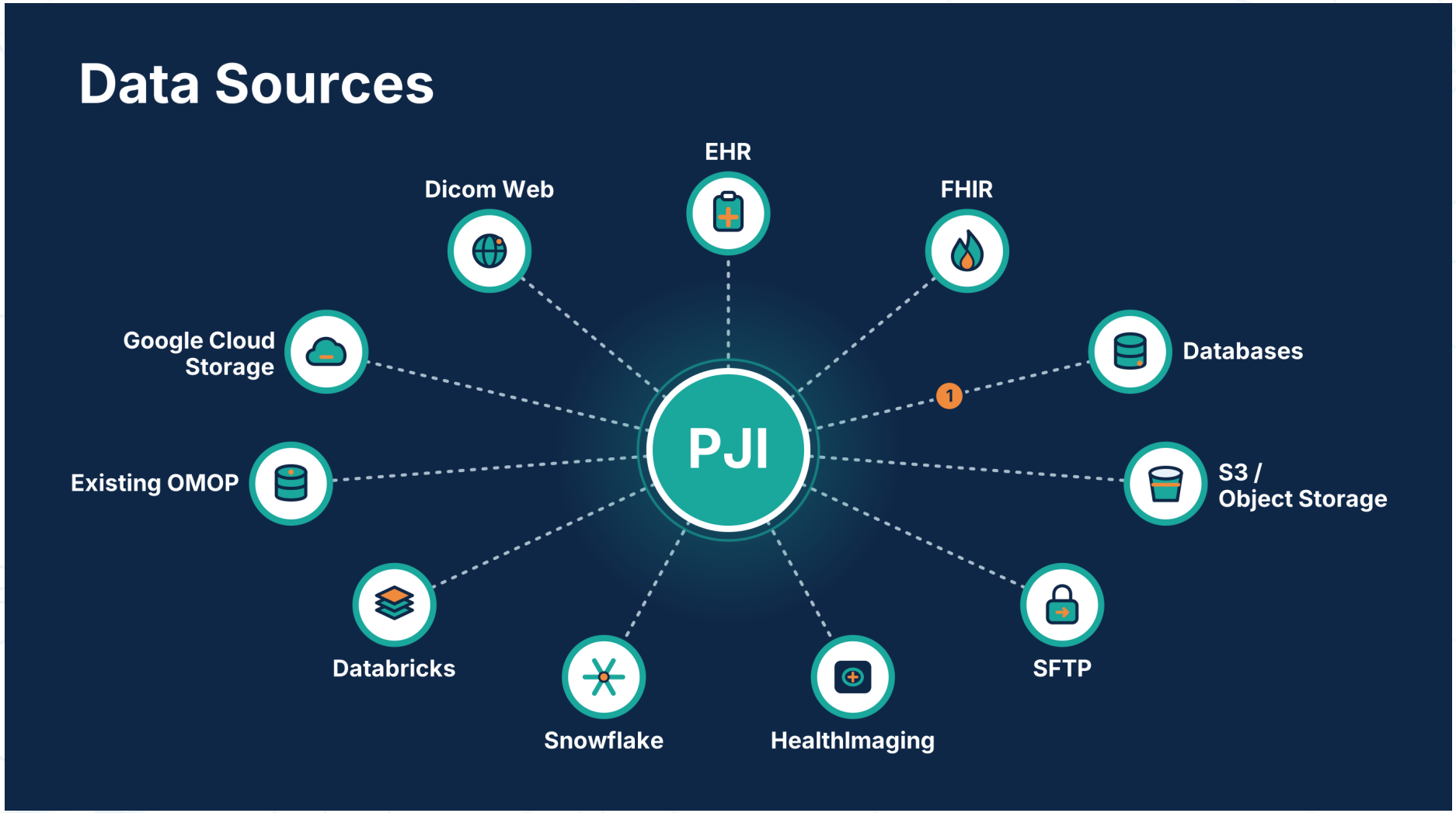
Gentle reminder
Still nothing gets sent to public internet or JSL side.



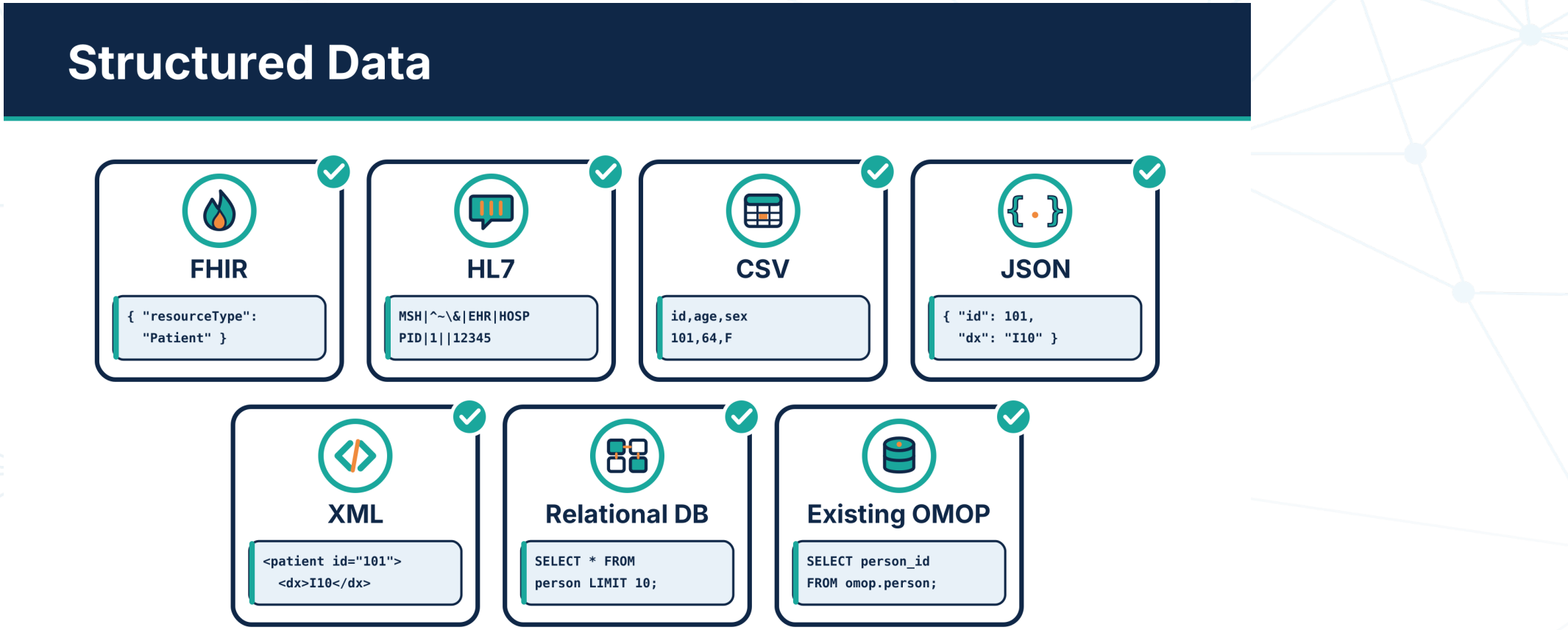
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Connect every source

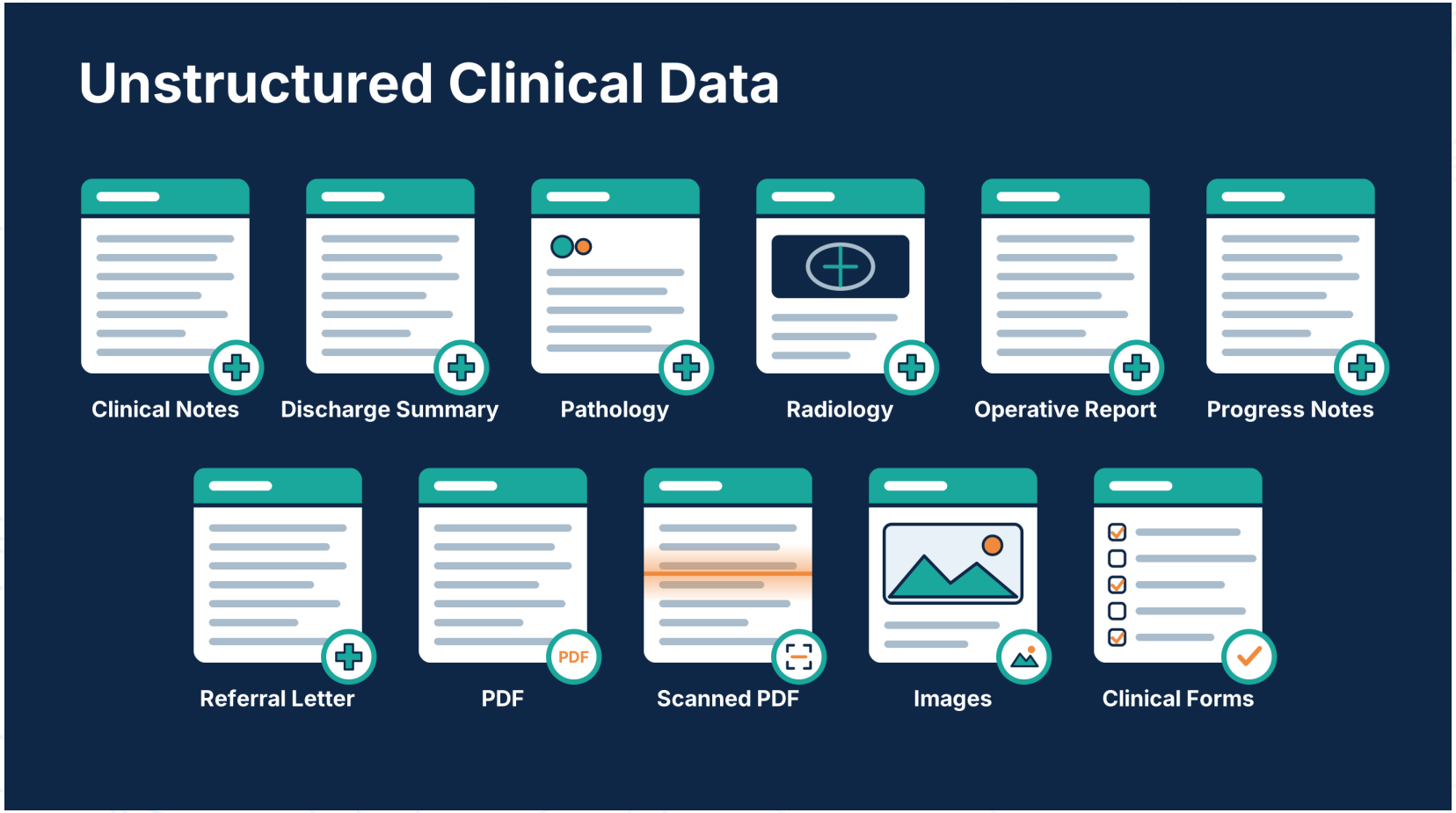


Structured Data Sources

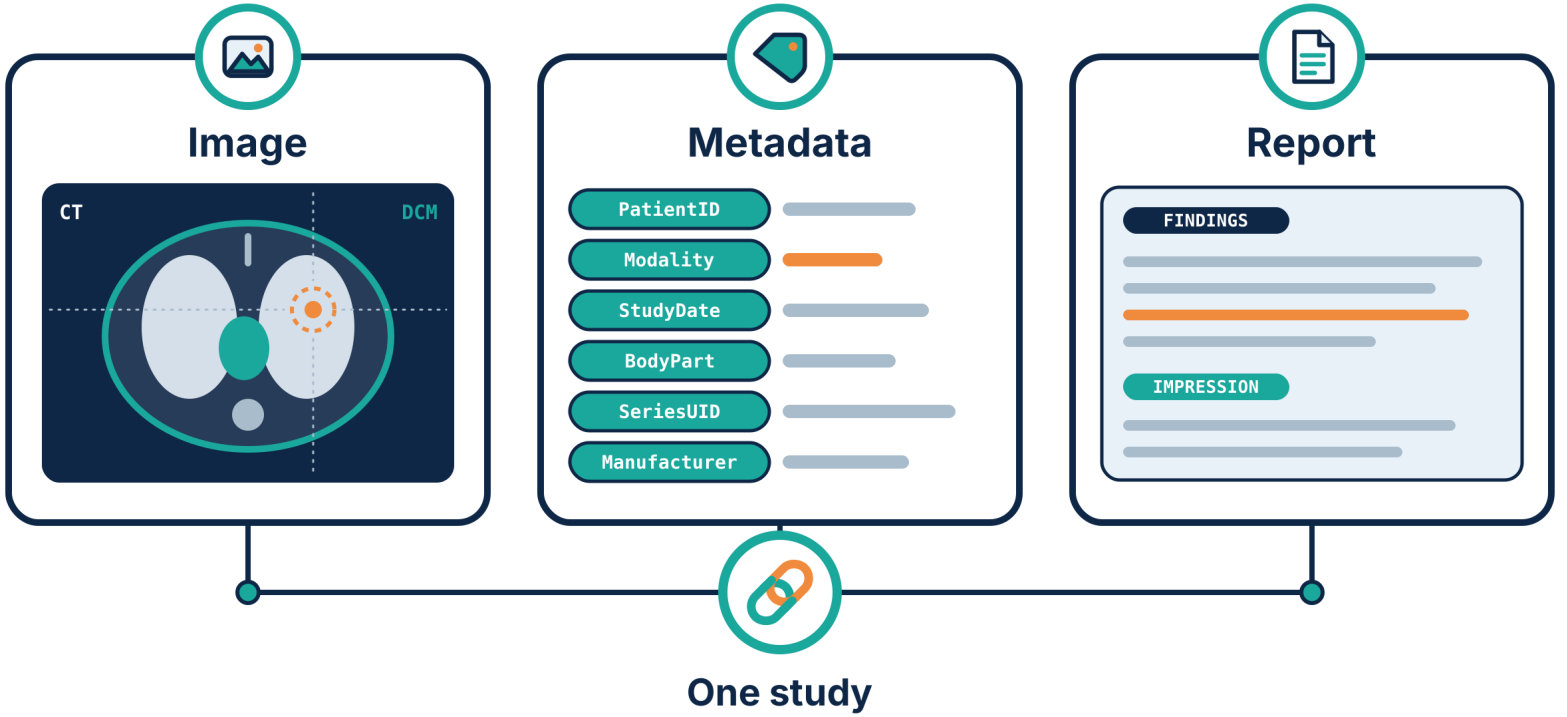


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Unstructured Clinical Data Types



Imaging: DICOM



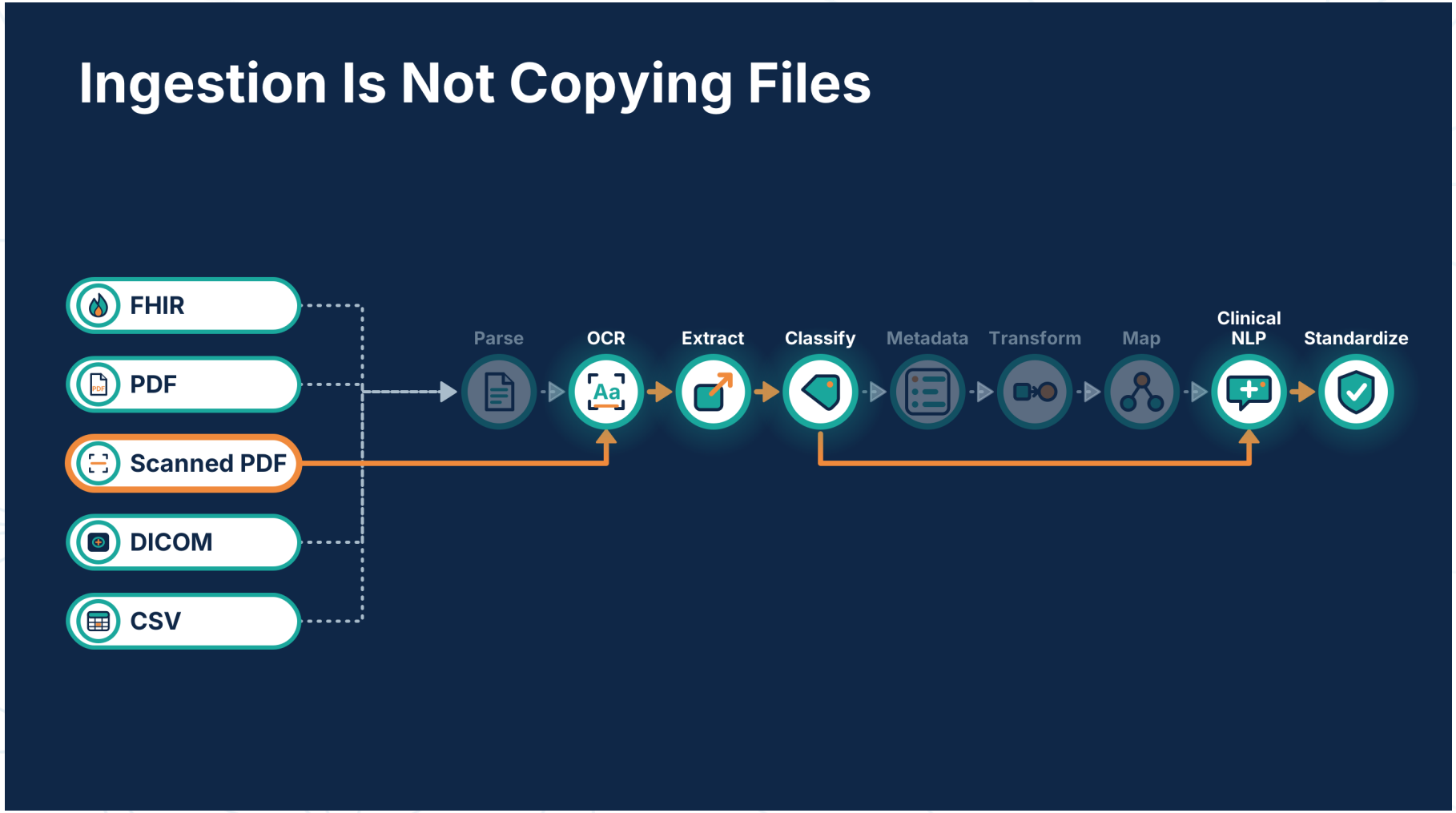
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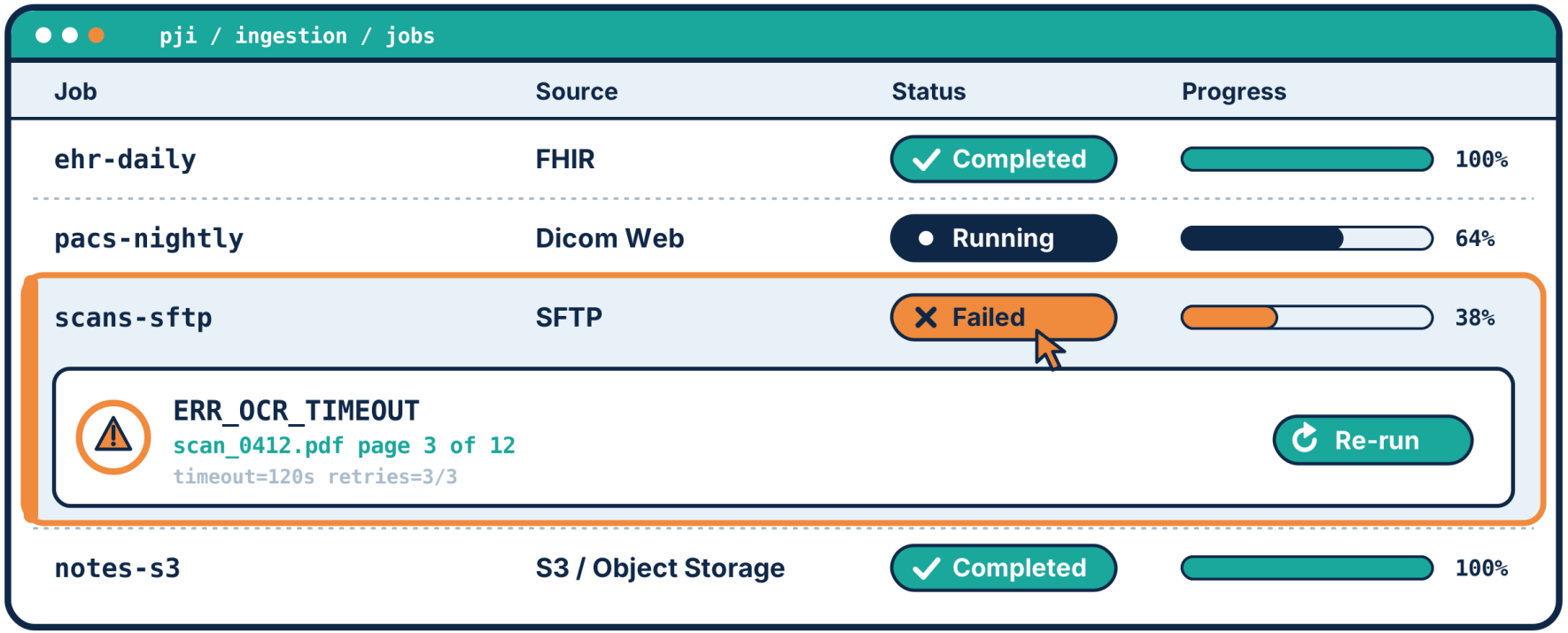
Ingestion jobs, more than copying files



Raw Data to Structured Form

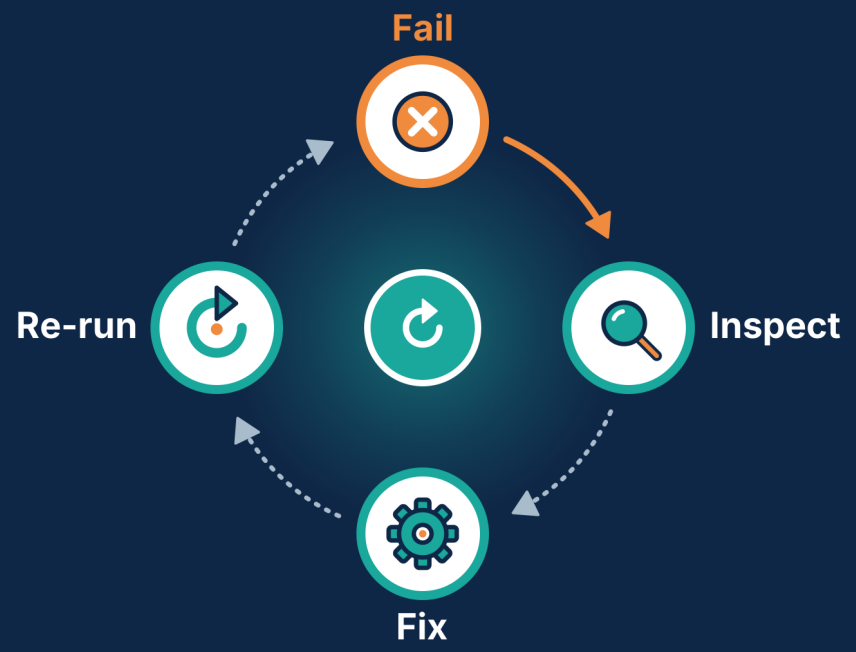


Monitoring Ingestion



Handling Ingestion Failure

Failures and Re-runs



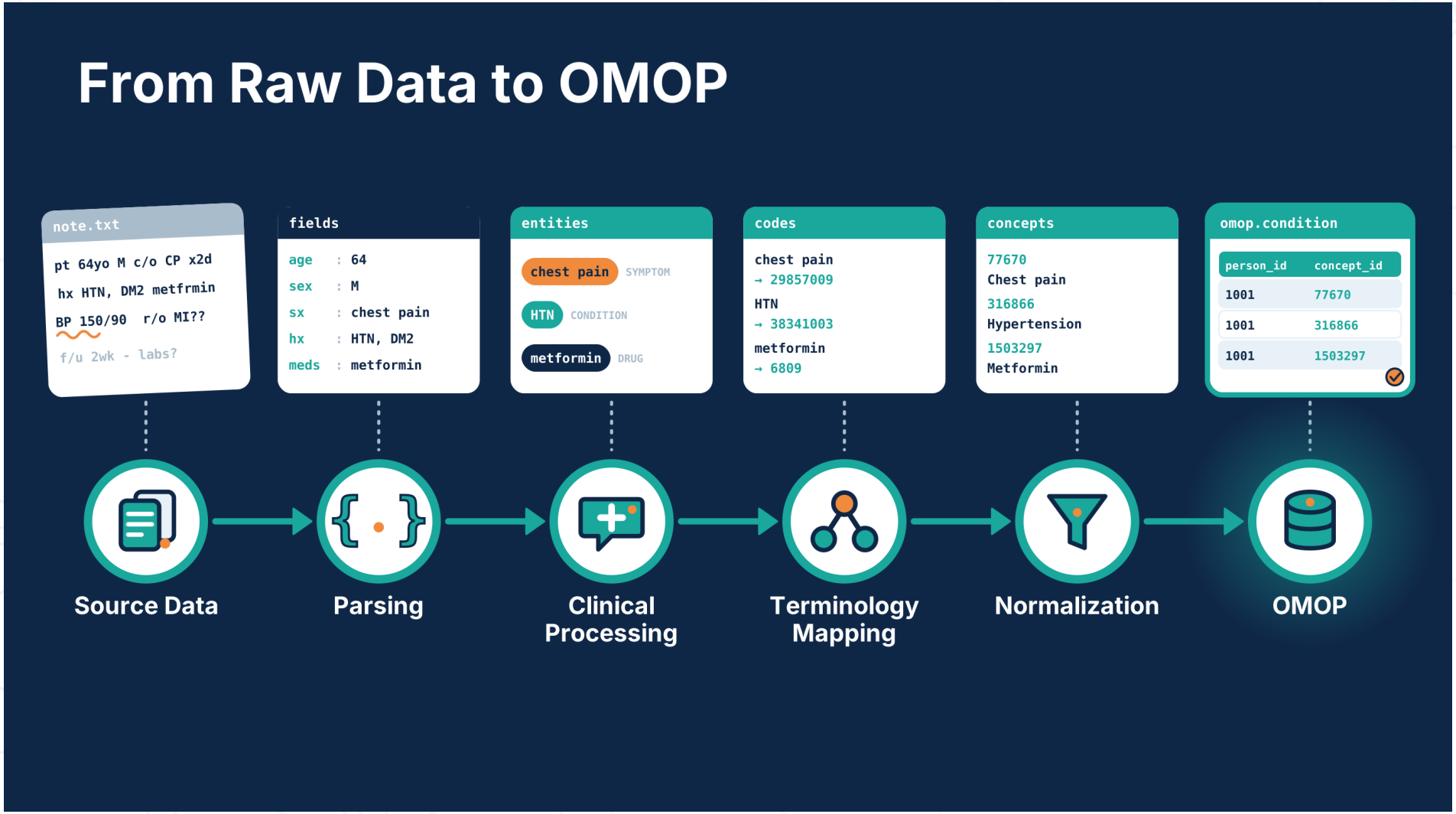


Recurring schedule

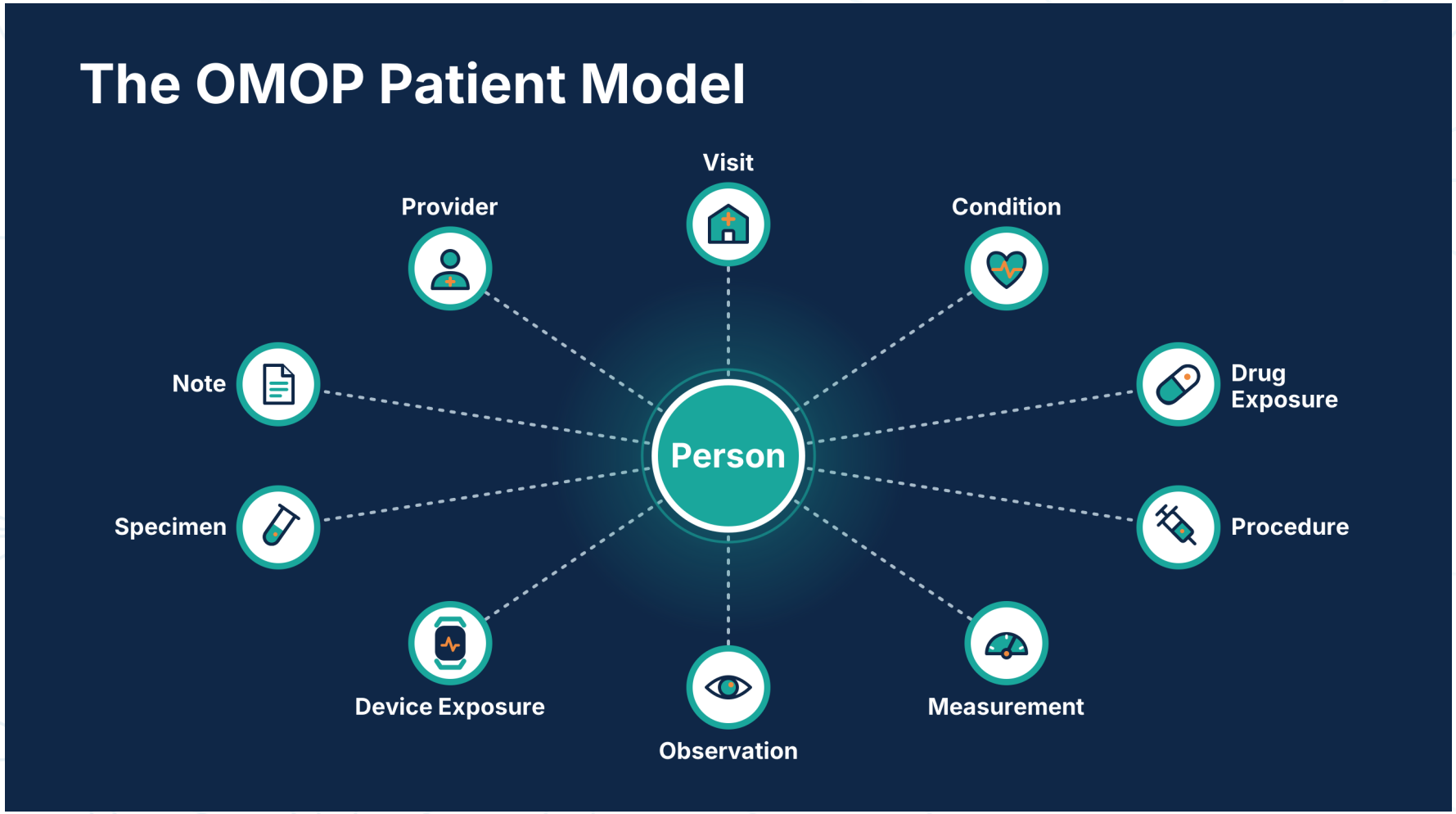
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Raw Data Transformation to OMOP Standard

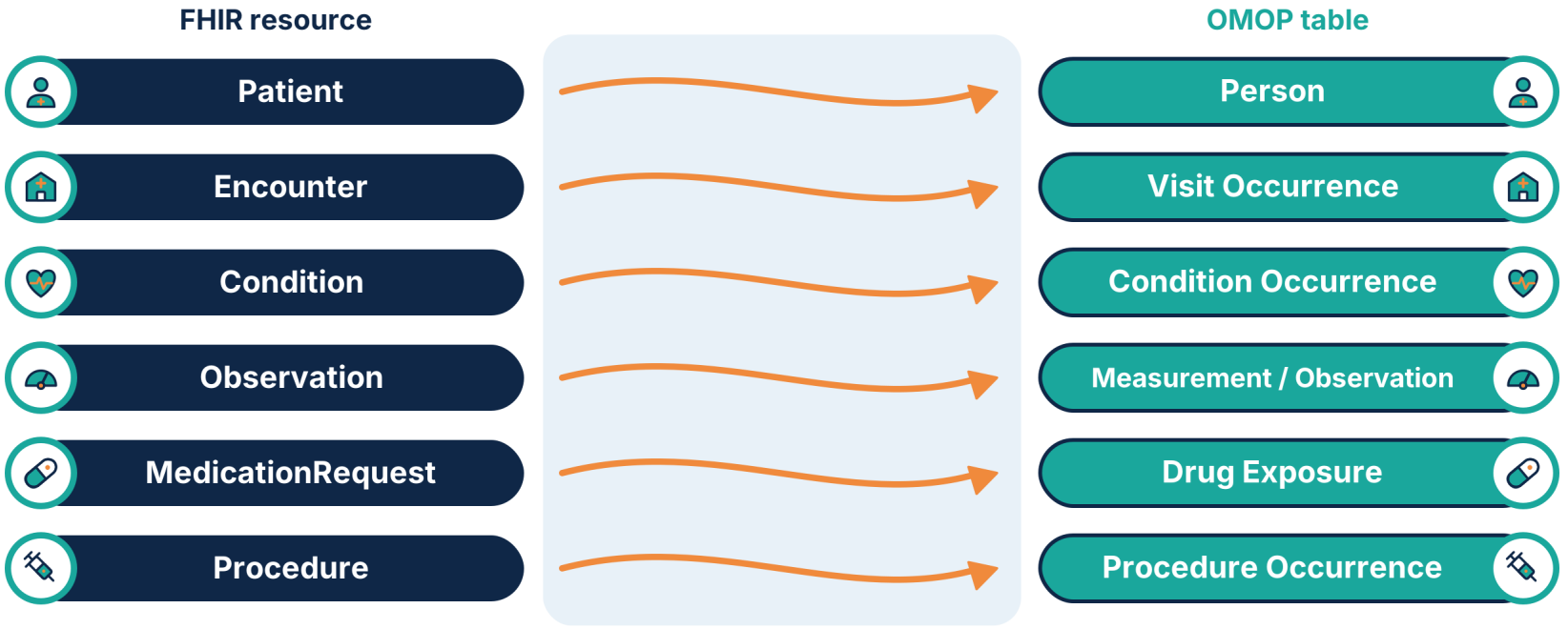


Standard OMOP Data Model



Conversion of FHIR to OMOP in PJI

FHIR → OMOP



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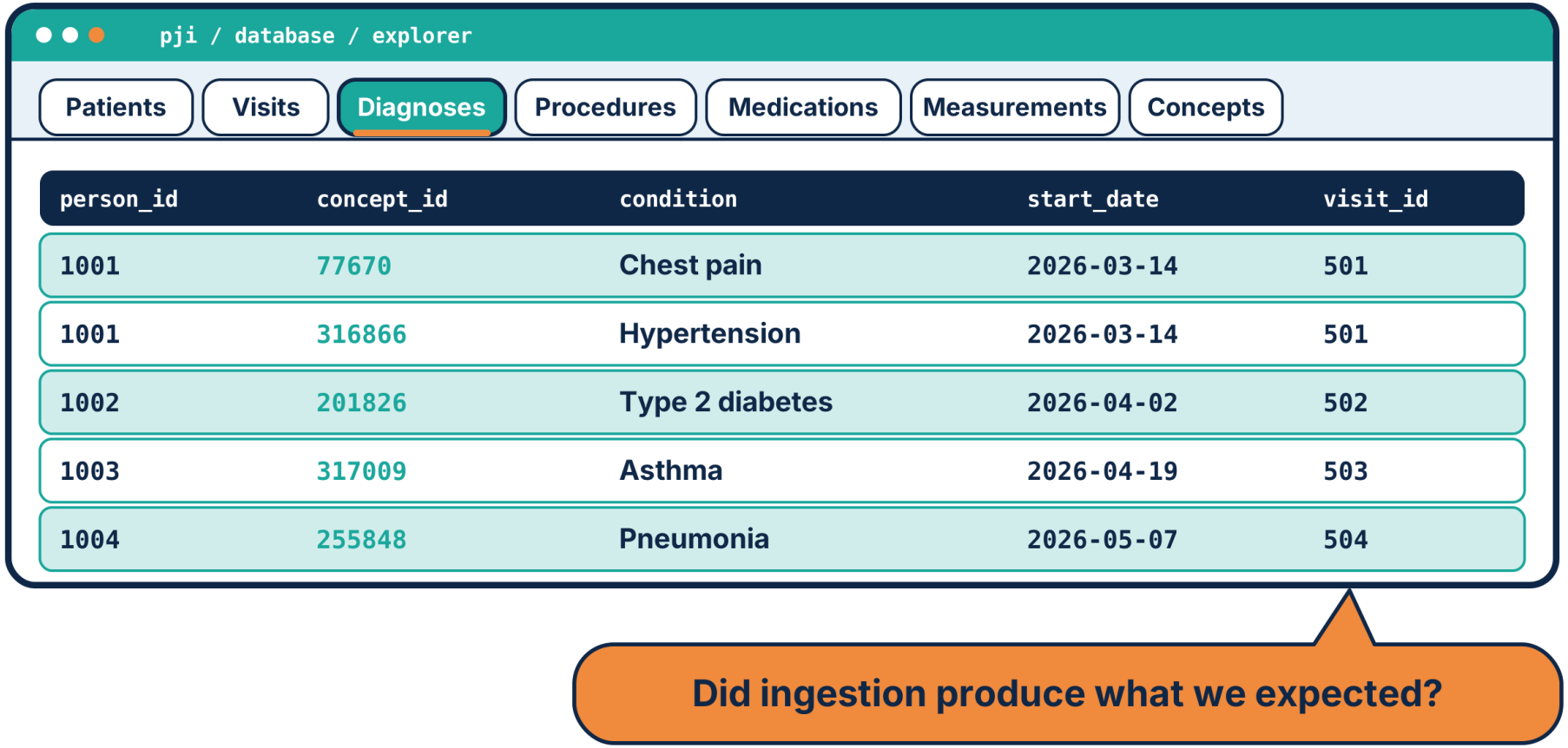
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Trust but verify with Database Explorer

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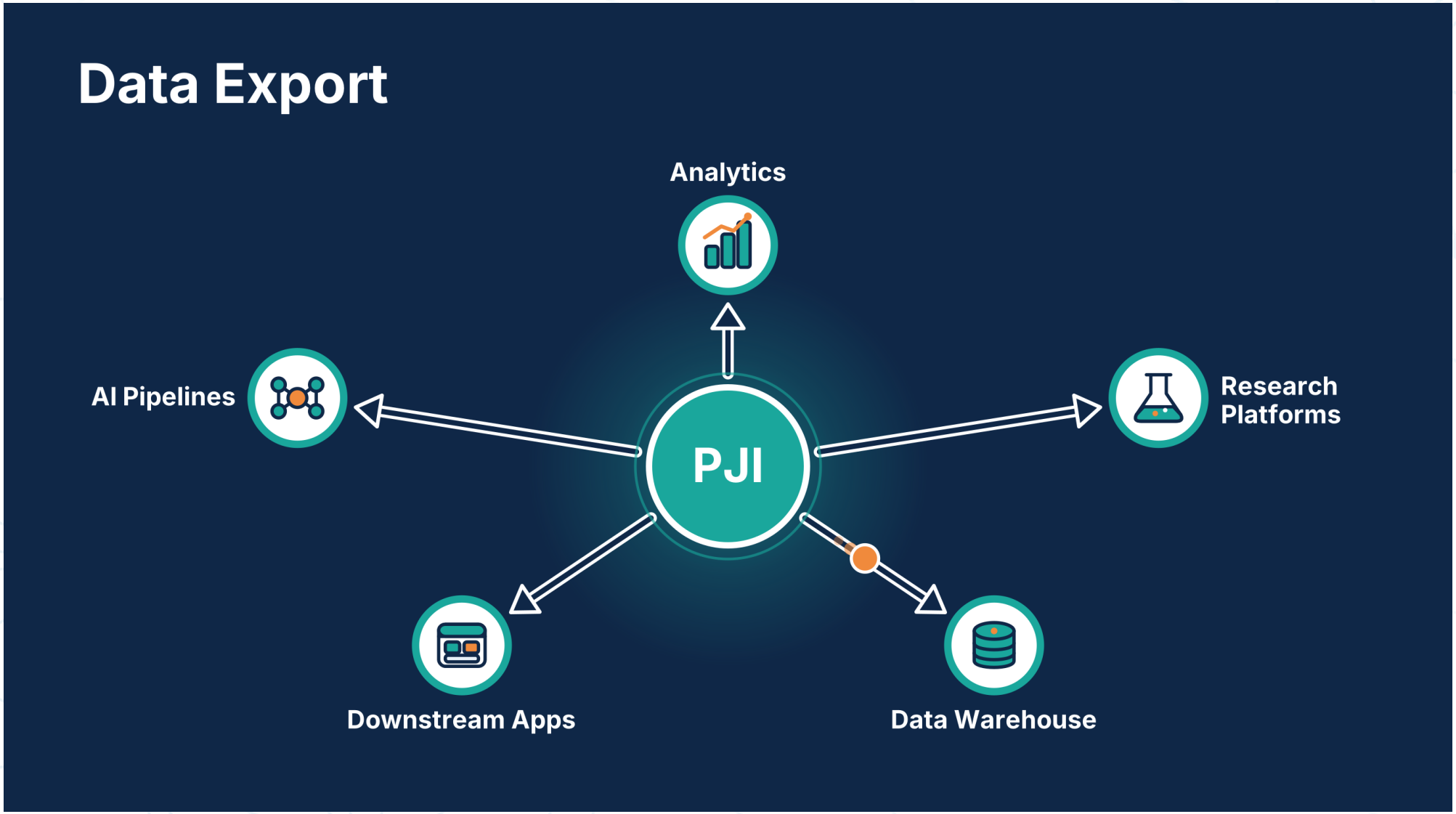
Database Explorer

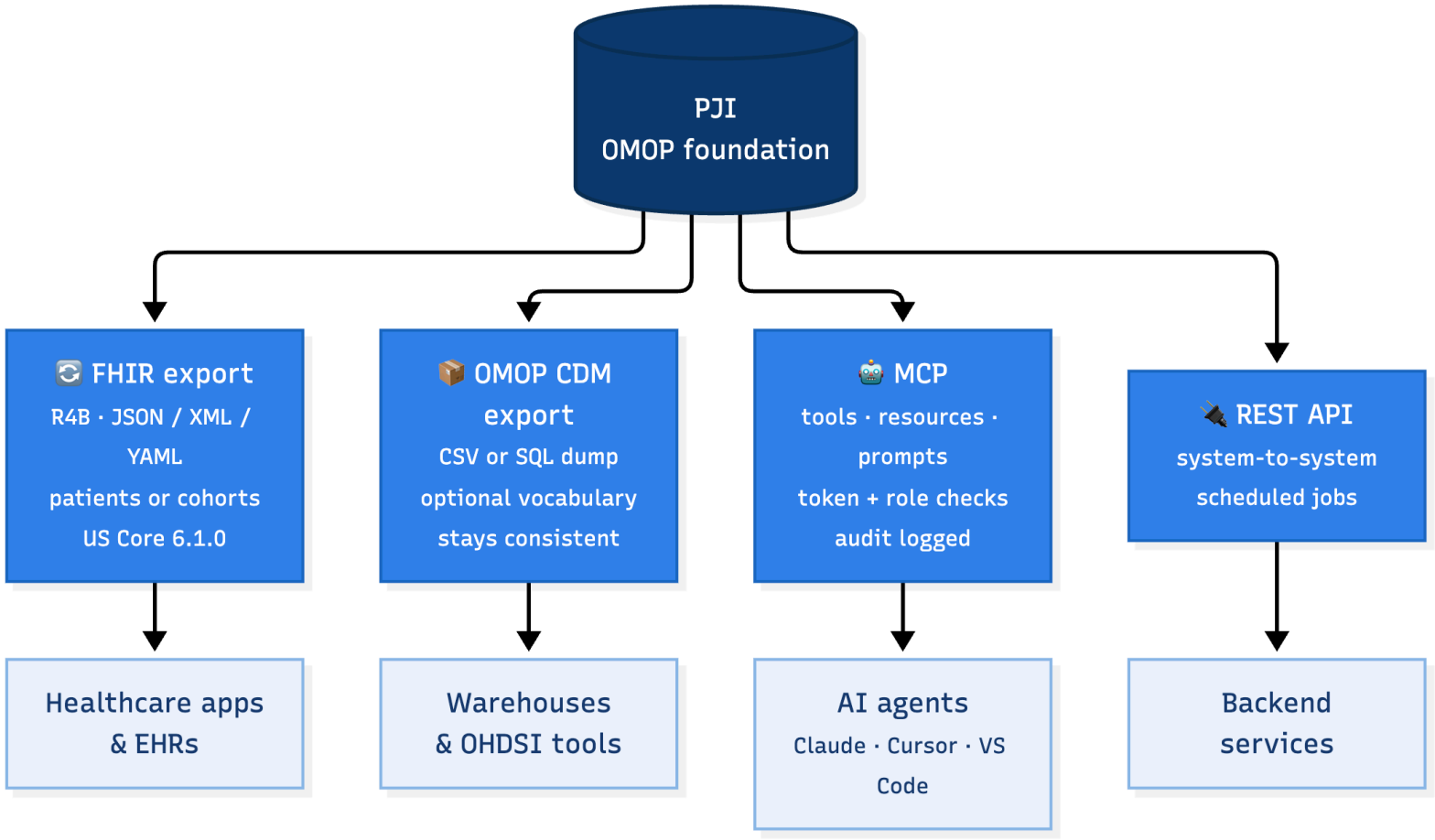


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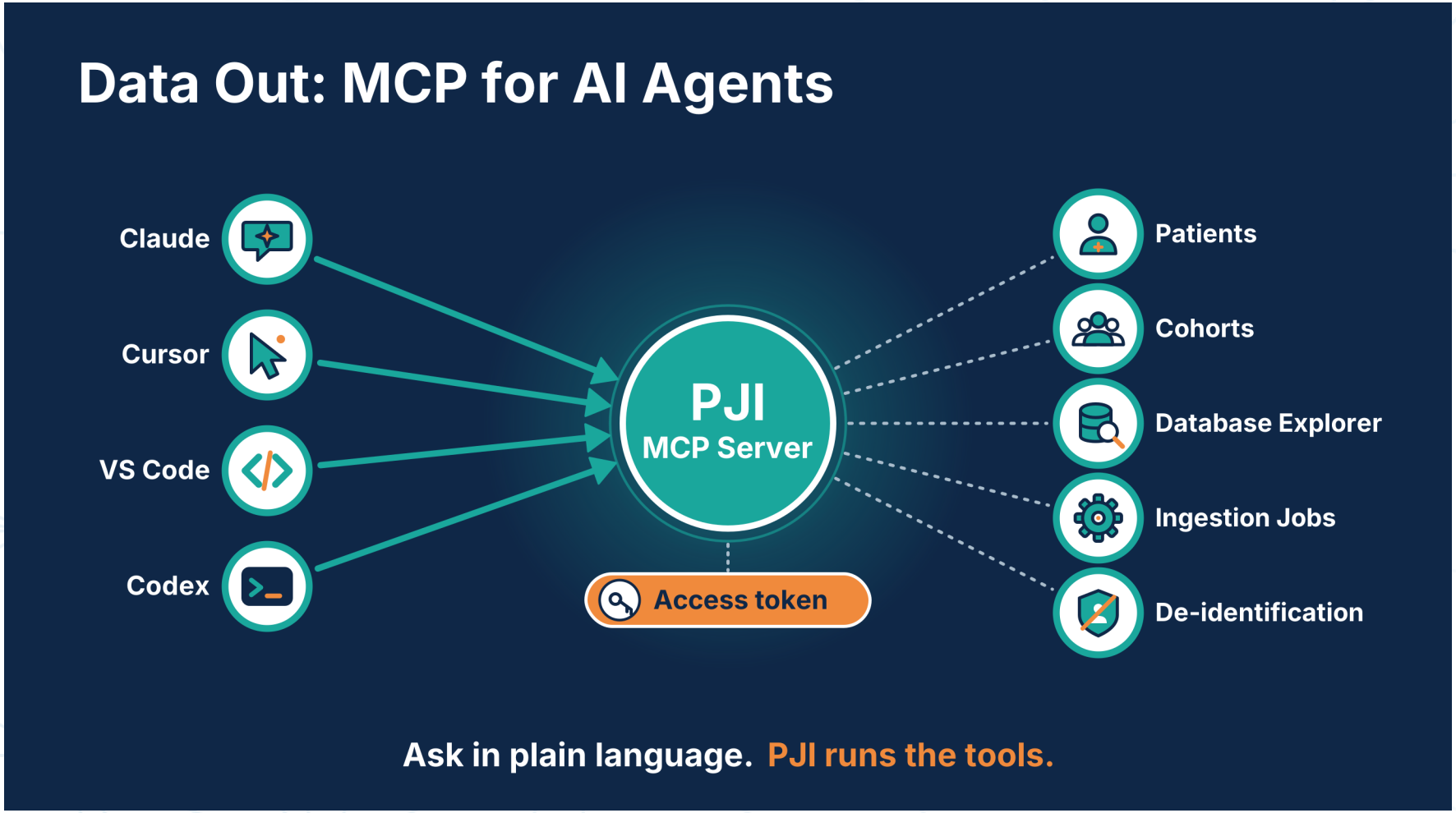
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Data out: export, FHIR and MCP




PJI
OMOP foundation
🔄 FHIR export
R4B · JSON / XML / YAML
patients or cohorts
US Core 6.1.0
📦 OMOP CDM export
CSV or SQL dump
optional vocabulary
stays consistent
🤖 MCP
tools · resources · prompts
token + role checks
audit logged
🔌 REST API
system-to-system
scheduled jobs
Healthcare apps & EHRs
Warehouses & OHDSI tools
AI agents
Claude · Cursor · VS Code
Backend services

Use PJI using AI agents



Patient Journey Intelligence

Your Data Never Leaves Home

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Deploy anywhere

Scale by adding nodes

Reuse existing compute

GPU only where needed

Pilot in days

AI agents via MCP

16+ sources

With support to ingest multiple modalities like scanned pdfs, DICOM images, text files, etc.

Major Providers

Enterprise-Grade Trust

protected by single sign-on, role-based access, write-only secrets and full audit logs.

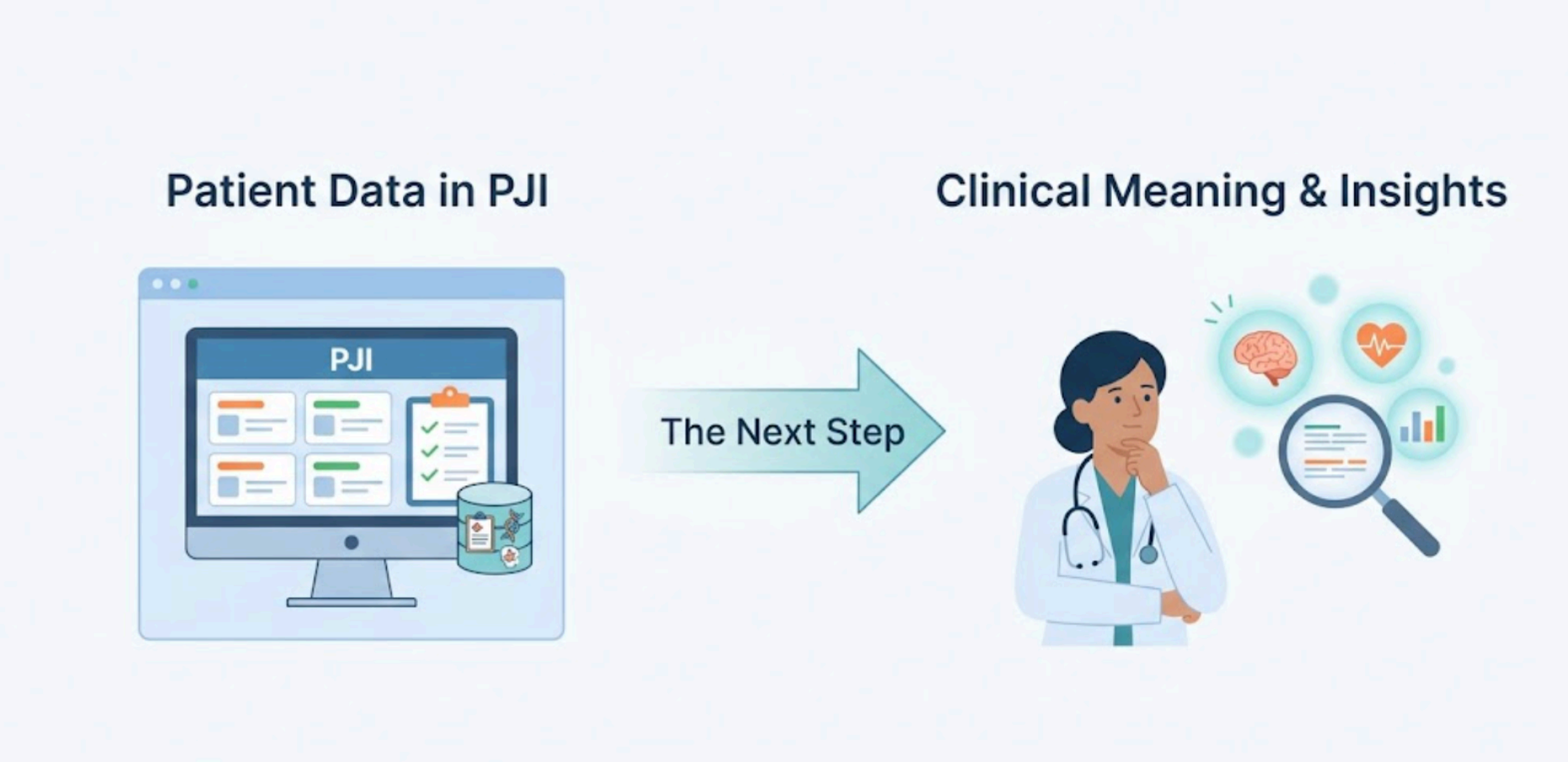
Your Data, Your Walls

10+ years

of real-world healthcare NLP experience behind this product.

Domain experts

What next?


Patient Data in PJI
PJI
The Next Step
Clinical Meaning & Insights

John Snow LABS

Thank you!

QUESTIONS?

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formational purposes only and represents the current view of John
itment on its part, and John Snow Labs cannot guarantee the